

3. Vinegar is the solution of which of the following?

- (a) Oxalic acid (b) Citric acid
(c) Hydrochloric acid (d) Acetic acid

42nd B.P.S.C. (Pre) 1997

Ans. (d)

Acetic acid (CH_3COOH) or ethanoic acid is an organic acid, which is responsible for sour taste and the pungent smell of the vinegar. The physical and chemical properties prove that the vinegar is a solution of acetic acid. It is produced from fermentation of ethanol.

4. The chemical name for vinegar is -

- (a) Methanol
(b) Ethanol
(c) Acetic acid
(d) Ethyl acetate
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (c)

Acetic acid is the chief component of vinegar. Vinegar is a liquid consisting mainly of acetic acid (CH_3COOH) and water. The acetic acid is produced by the fermentation of ethanol (formed from sugarcane juice and molasses) by acetic acid bacteria. It is mainly used as cooking ingredient and as a preservative.

5. The odor of acetic acid resembles that of

- (a) vinegar (b) tomato
(c) kerosene (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

The Odor of acetic acid, especially when ions are in its concentrated form, resembles that of vinegar. Vinegar is a diluted solution of acetic acid in water and it is widely used in cooking food preservation and various household applications.

6. Which acid is described as HOOCCOOH ?

- (a) Carbonic acid (b) Oxalic acid
(c) Acetic acid (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

The acid described as HOOCCOOH is oxalic acid. Its chemical formula is $\text{C}_2\text{H}_2\text{O}_4$. Oxalic acid is a dicarboxylic acid with two carboxylic acid functional Group (COOH) and is commonly found in certain plant like spinach, shubarb, and others. It is also use as a cleaning agent and in some industrial processes.

7. Which of the following is an acid-base indicator?

- (a) Vinegar (b) Turmeric
(c) Baking soda (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

In chemistry, indicators are chemical compounds that indicate the presence or absence of another compound by a sign of visual change like a change of color of the solution. Among the given options, turmeric changes its color to reddish brown in basic solution but in the case of the acid solution, the color of turmeric remains the same. So it can be used as an acid-base indicator.

8. Which is the natural source of acetic acid?

- (a) Tomato (b) Orange
(c) Vinegar (d) Tamarind

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (c)

Vinegar is natural source of acetic acid. Vinegar is a combination of acetic acid and water made by a two-step fermentation process.

E. The Explosive Substances

1. The quality of petrol is expressed by :

- (a) Cetane number (b) Octane number
(c) Butane number (d) Reynolds number
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (b)

The octane number of a fuel (petrol) is the measure of its antiknock quality when it is used in a spark ignition, internal combustion engine as compared to the antiknock quality of ISO-octane. The higher octane number of a fuel means the better anti-knocking capacity for that fuel.

2. Cooking gas is mainly –

- (a) Carbon dioxide (b) Carbon monoxide
(c) Methane (d) Nitrogen and Oxygen

44th B.P.S.C. (Pre) 2000

Ans. (c)

Methane is the chief component of biogas or gobar gas which is used for domestic purposes.

3. Which among the following is the main constituent of biogas?

- (a) Methane (b) Propane

- (c) Butane (d) Ethane
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (a)

When the gas is produced by the biological breakdown of an organic material in the absence of oxygen, it is called biogas. Methane and carbon dioxide are the main constituents of the biogas. It may contain small quantity of hydrogen sulfide, moisture and siloxane. Dung gas or biogas contains methane up to 50-75 percent.

4. LPG used as domestic fuel mainly contains

- (a) Methane (b) Propane
(c) Ethylene (d) Butane
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

LPG is the abbreviation or short form for Liquefied Petroleum Gas. Like all fossil fuels, it is a non-renewable source of energy. It is extracted from crude oil and natural gas. The main composition of LPG are hydrocarbons containing three or four carbon atoms. The normal components of LPG are propane (C_3H_8) and butane (C_4H_{10}). LPG mixtures contain a little more propane in winter and a little more butane during summer but, on average, it is approximately 65% butane and 35% propane. Small concentrations of other hydrocarbons like ethane may also be present. Ethyl mercaptan is normally used as a stenching agent for smell.

5. The Pressure gauge is not given in the domestic LPG cylinders because-

- (a) They are so costly
(b) They cannot represent the quantity of gas in LPG cylinders
(c) Their use is not safe
(d) They are choked by LPGs

42nd B.P.S.C. (Pre) 1997

Ans. (b)

Oxygen and other gases found in the gas cylinders are in a gaseous state while in domestic LPG cylinders, gas is found in the liquid state so pressure gauge cannot represent the quantity of gas in LPG cylinders. That is why the pressure gauge is not given in the domestic LPG cylinders.

6. Which of the following is the major constituent of liquefied petroleum gas?

- (a) Methane (b) ethane
(c) propane (d) butane
(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (e)

The main components of liquefied petroleum gas (LPG) are butane (about 65%) and propane (about 35%). It may also contain small amounts of other hydrocarbons such as ethane.

7. The chief source of aromatic compound is

- (a) fruits (b) oil of wintergreen
(c) petroleum (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 07-12-2023

Ans. (c)

The chief source of aromatic compound is petroleum. Aromatic compounds, originally named because of their fragrant properties, are unsaturated hydrocarbon ring structures that exhibit special properties, including unusual stability, due to their atomicity. These compounds are generally non-polar and immiscible with water.

8. The major component of CNG is :

- (a) CO_2 (b) N_2
(c) H_2 (d) CH_4
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)

CNG (compressed natural gas) is mainly composed of methane (CH_4). It is a mixture of hydrocarbons which is found deep inside Earth and have approx. 80% to 90% of methane.

9. Oxygen is absent in -

- (a) Kerosene (b) Glass
(c) Soil (d) Cement
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

65th B.P.S.C. (Pre) 2019

Ans. (a)

Kerosene is a combustible hydrocarbon liquid which is derived from petroleum. It is a mixture of hydrocarbons and does not contain oxygen. Oxygen is present in glass, soil and cement.

10. A form of matter has no fixed shape but it has a fixed volume. An example of this form of matter is

- (a) carbon steel (b) krypton
(c) kerosene (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

The state of matter which has no definite shape but has a definite or fixed volume is called liquid. Kerosene is a substance that is present in a liquid state thus, it has a definite volume but no actual shape.